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SPCE 5085
Assignment #4

7.2 — Why is attenuation on a slant path in clear sky conditions greater at 22 GHz than at 30 GHz? Why is the occurrence of attenuation in rain on a slant path at the 1 - 2 dB level often greater at 22 GHz than at 30 GHz?

There is a peak of attenuation at 22 GHz because of a property of water molecules. This effect is strongest at 22 GHz and weaker at 30 GHz. This effect is still dominant when attenuation is at the 1 -2 dB level, because the attenuation contributed by the molecules is greater than the attenuation contributed by rain.

7.4 — An earth station at sea level communicates at an elevation angle of 30∘with a GEO satellite in Ka-band at a downlink frequency of 20 GHz. Stratiform rain is present in the slant path with a melting level height of 2.5 km. Find

a The path length through the rain.

Since the earth station is at sea level, the rain height above the station equals the melting layer height, $h = 2.5$ km. The slant path length through the rain is

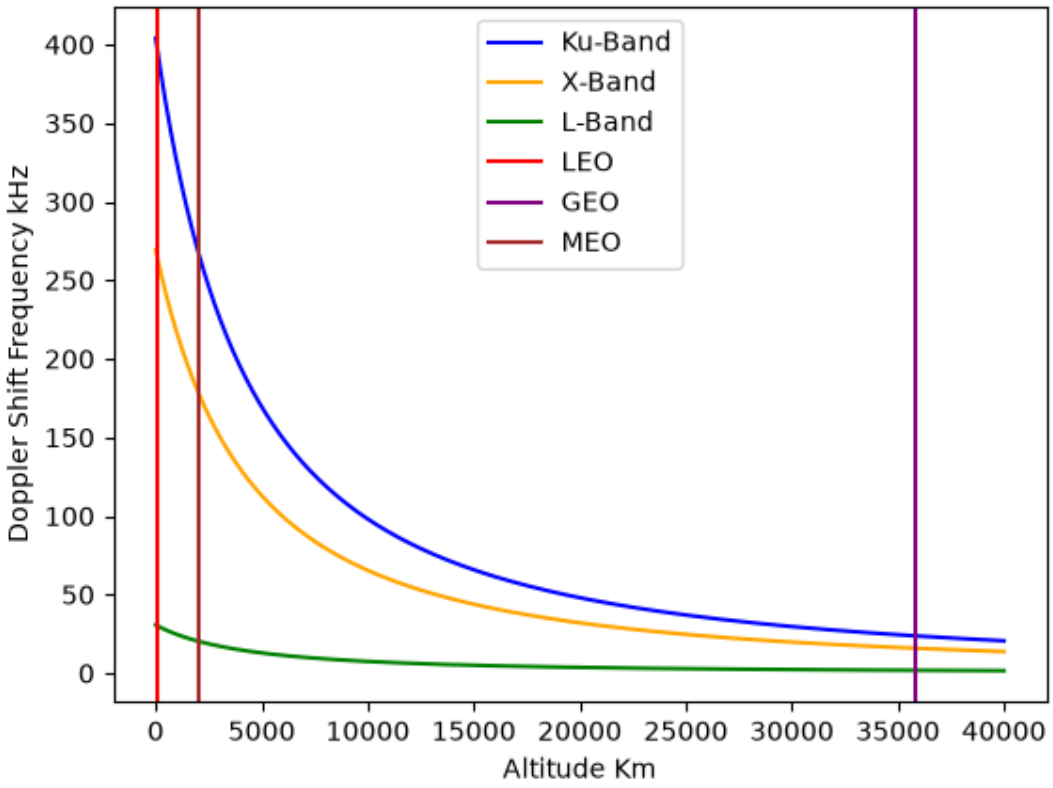
$$L = \frac{h}{\sin(\theta)} = \frac{2.5 \text{ km}}{\sin(30\circ)} = \frac{2.5 \text{ km}}{0.5} = \boxed{5.00 \text{ km}}$$

b The path attenuation for a specific attenuation of 1.5 dB/km.

The path attenuation is the product of the specific attenuation and the path length through the rain found in part (a):

$$A = \gamma L = (1.5 \text{ dB/km})(5.00 \text{ km}) = \boxed{7.50 \text{ dB}}$$

3 — Create a plot similar to figure 3.1 Doppler vs SV altitude in the article "The Mathematical Model of Doppler Frequency Shift in Leo At Ku, K and Ka Frequency Bands" (posted in the module 2), for satellite signal transmission from spacecraft to stationary ground station - L band (1.25 GHz), X band (11.0 GHz), and Ku band (16.5 GHz) for SV altitudes of (at a minimum) 1000, 10,000 and 20,000 km. (40 pts)



Source:
https://github.com/chrismcarrico/homework/blob/master/spce5085/a4/carrico_spce5085_a4.ipynb